Instant Forgetting: Steering Vectors Leave No Trace in Transformer Hidden States

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Abstract

Activation steering—adding learned direction vectors to model hidden states—is increasingly used to control language model behavior at inference time. Standard practice applies steering vectors at every token position, but the temporal dynamics of steering influence remain poorly understood: if a vector is applied briefly and then removed, how quickly does its effect on hidden states dissipate? We present the first direct empirical characterization of post-removal steering vector decay. Using GPT-2-XL and contrastive activation addition across three behavioral dimensions (truthfulness, sentiment, formality), we measure token-by-token cosine similarity and KL divergence after steering removal. Our central finding is that representational decay is *instantaneous*: within a single generation step, hidden states at the steered layer revert to values indistinguishable from an unsteered model (Δ cosine similarity drops from $\sim$0.019 to $\sim$0.000). This result holds across all layers, multiplier strengths, and steering durations tested. Crucially, autoregressive and teacher-forced conditions produce *identical* hidden states (Cohen's $d = 0.000$), demonstrating that the model retains no hidden state memory of the steering intervention. Behavioral effects persist solely through the tokens generated under steering, which remain in the context and influence subsequent generation. These findings establish that continuous vector application is necessary for sustained representational steering and that the autoregressive token sequence, not hidden state momentum, is the primary carrier of steering influence.

1 Introduction

Activation steering has emerged as a lightweight method for controlling language model behavior without fine-tuning [Turner et al., 2023, Panickssery et al., 2024, Zou et al., 2023]. By adding a learned direction vector to a model's hidden states during the forward pass, practitioners can increase truthfulness, reduce toxicity, or shift stylistic register. The approach is attractive because it requires no gradient updates, operates at inference time, and needs only a small set of contrastive examples to extract a steering direction.

A natural question arises: must steering vectors be applied at every token position, or can brief application produce lasting effects? Standard methods such as Contrastive Activation Addition [Panickssery et al., 2024] inject vectors continuously, implicitly assuming that the effect would not persist without reinforcement. Yet no prior work has directly measured how quickly steering influence dissipates after the vector is removed. Understanding this decay is important for three reasons. First, practitioners want to minimize computational overhead by applying vectors only when needed. Second, the decay rate reveals how the model processes and absorbs perturbations to its internal representations. Third, the distinction between representational decay (hidden states reverting to their natural trajectory) and behavioral persistence (generated tokens maintaining influence through the context) has direct implications for steering method design.

We provide the first empirical characterization of post-removal steering vector decay. Using GPT-2-XL (1.5B parameters) and TransformerLens [Nanda and Bloom, 2022], we extract steering vectors for three behavioral dimensions—truthfulness, positive sentiment, and formal style—via contrastive

activation addition. We then apply each vector for the first K tokens of generation, remove it, and measure hidden state dynamics token-by-token for 20 subsequent tokens. We compare six conditions: no steering, continuous steering, autoregressive generation after removal (AR), teacher-forced generation with the same steered tokens (TF), teacher-forced generation with unsteered tokens (TF-CLEAN), and a random-vector control.

Our central finding is that representational decay is *instantaneous*. The cosine similarity delta between steered and unsteered hidden states drops from $\sim$0.019 to $\sim$0.000 within a single generation step after steering removal. This result is consistent across all three behaviors, all layers tested (8, 16, 24, 32), all multiplier strengths (1–5$\times$), and all steering durations (K=1 to K=10). Autoregressive and teacher-forced conditions produce identical hidden states (Cohen's $d = 0.000$), confirming that the model retains no hidden state memory of the perturbation. Yet behavioral effects persist: the KL divergence between the steered model's output distribution and the unsteered baseline remains elevated for many tokens, with a half-life exceeding 75 tokens (figure 2). This persistence vanishes entirely under the TF-CLEAN condition, where unsteered tokens replace the steered context, confirming that behavioral influence is carried by the generated token sequence rather than by hidden state dynamics.

We make the following contributions:

- We present the first token-by-token measurement of steering vector decay dynamics, showing that representational influence is absorbed within a single forward pass.
- We decompose steering persistence into representational and behavioral components, demonstrating that behavioral effects operate entirely through the autoregressive token feedback loop.
- We conduct comprehensive ablations across layers, multiplier strengths, steering durations, and behavioral dimensions, establishing the universality of instantaneous representational decay.

2 Related Work

Activation steering. The linear representation hypothesis [Park et al., 2023, Marks and Tegmark, 2023] posits that high-level concepts correspond to approximately linear directions in transformer activation space. ACTADD [Turner et al., 2023] exploits this by computing a steering vector from the activation difference between a positive and negative prompt pair and adding it at a single layer during the forward pass. CAA [Panickssery et al., 2024] extends this to use mean-difference vectors from hundreds of contrastive pairs, applied at all token positions after the instruction. The design choice of continuous injection implicitly acknowledges that single-application steering would decay, though neither work measures this directly. Representation Engineering [Zou et al., 2023] provides a broader framework for reading and writing model representations, treating steering as manipulating a learned representation space.

Evidence for steering decay. Several papers provide indirect evidence that steering effects diminish during generation. Li et al. [2026] demonstrate that static steering vectors become misaligned during autoregressive generation because hidden states drift through activation space, causing reliability to degrade in long-form generation. They propose Steering Vector Fields (SVF), which re-compute the steering direction from the current hidden state every K steps. Nguyen et al. [2025] formalize steering as a control theory problem and prove that proportional-only control (equivalent to standard activation addition) has inherent steady-state error. Their PID controller adds integral and derivative terms to counteract this error. Bigelow et al. [2025] model steering through a Bayesian lens, where the steering vector shifts a concept prior that can be overridden by evidence accumulated during generation. Unlike these works, which propose solutions to decay, we directly measure the decay dynamics themselves.

Mechanisms of decay. Several mechanisms have been proposed for how perturbations are absorbed. Xu et al. [2026] argue that model activations lie on a low-dimensional manifold, and subsequent layers project perturbed activations back onto this manifold. Dang and Ngo [2026] identify norm distortion as a specific mechanism: adding a steering vector violates activation norms expected by LayerNorm, causing distribution shift that compounds across layers. They propose norm-preserving rotation as an alternative. The Hidden Life of Tokens [Ramesh et al., 2025] shows that information injected into hidden states diminishes as more tokens are generated, suggesting that perturbations are naturally attenuated by the autoregressive process.

Behavioral persistence and error propagation. Bachmann and Nagarajan [2024] establish that small perturbations in hidden states can compound through autoregressive generation due to the teacher-forcing gap. This suggests competing effects: the steering perturbation may decay in hidden states, but behavioral changes caused by early steered tokens may persist as the model conditions on its own outputs. Our experimental design explicitly separates these two effects by comparing autoregressive generation (where both operate) with teacher-forced conditions (where only representational decay is measured).

Counteracting decay. Beyond continuous injection [Panickssery et al., 2024], methods include context-dependent refreshing [Li et al., 2026], PID control [Nguyen et al., 2025], conditional steering [Lee et al., 2024], semantics-adaptive intervention [Wang et al., 2024], norm-preserving rotation [Dang and Ngo, 2026], SAE-targeted steering [Chalnev et al., 2024], and KV cache steering [Zhang et al., 2025]. Our work complements these by characterizing the problem they address: we show that without continuous reinforcement, steering influence is lost within a single step, motivating the need for these approaches.

Evaluation of steering. Pres et al. [2024] demonstrate that multiple-choice evaluations can overstate steering success and that long-form generation is a stricter test. Tan et al. [2024] and Bas et al. [2025] show that steerability varies across concepts and follows an inverted-U curve with steering strength. These findings motivate our choice to evaluate across multiple behaviors and to measure both short-term (hidden state) and long-term (output distribution) effects.

3 Methodology

We describe our experimental setup for measuring steering vector decay dynamics, including model selection, steering vector extraction, experimental conditions, and evaluation metrics.

3.1 Model and Infrastructure

We use GPT-2-XL (1.5B parameters, 48 layers, $d_{\text{model}} = 1600$), loaded via TransformerLens [Nanda and Bloom, 2022] in float16 on an NVIDIA RTX A6000 GPU (49 GB VRAM). We select GPT-2-XL because it is well-supported by TransformerLens, fits on a single GPU, and was used in the original ActAdd paper [Turner et al., 2023]. All generation uses greedy decoding (argmax) with a fixed random seed of 42 for reproducibility.

3.2 Steering Vector Extraction

We construct contrastive pairs for three behavioral dimensions: truthfulness (20 pairs of true vs. false factual statements), positive sentiment (20 pairs of positive vs. negative evaluative language), and formal style (20 pairs of formal vs. informal register). Each pair consists of semantically similar sentences differing primarily in the target dimension (e.g., "The Earth revolves around the Sun" vs. "The Sun revolves around the Earth" for truthfulness).

For each behavior and target layer ℓ, we compute the mean activation of the last token across all positive and negative examples, then take the normalized difference:

$$\boldsymbol{v}_{\text{steer}}^{(\ell)} = \frac{\bar{\boldsymbol{h}}_{(\ell)}^{+} - \bar{\boldsymbol{h}}_{(\ell)}^{-}}{\|\bar{\boldsymbol{h}}_{(\ell)}^{+} - \bar{\boldsymbol{h}}_{(\ell)}^{-}\|} \tag{1}$$

where $\bar{\boldsymbol{h}}_{(\ell)}^{+}$ and $\bar{\boldsymbol{h}}_{(\ell)}^{-}$ are the mean activations at layer ℓ for the positive and negative examples, respectively.

3.3 Experimental Conditions

We generate continuations from 25 diverse open-ended prompts (e.g., "The most important thing to remember is that"), designed to be neutral with respect to all behavioral dimensions. For each prompt, we run six conditions:

1. **No Steering** (baseline): Normal generation with no vector applied.

2. **Continuous Steering**: Vector $\alpha \cdot \boldsymbol{v}_{\text{steer}}$ added at every token position, where α is the multiplier strength.
3. **AR** (Autoregressive): Vector applied for the first K tokens, then free generation without the vector.
4. **TF** (Teacher-Forced, steered tokens): Vector applied for K tokens, then the same tokens that AR generated are fed as input, but without the vector.
5. **TF-Clean** (Teacher-Forced, clean tokens): Vector applied for K tokens, then tokens from the unsteered baseline are fed without the vector.
6. **Random** (control): A random unit vector with the same multiplier applied for K tokens, then free generation.

The TF condition tests whether the model retains any hidden state memory of the steering intervention beyond what the generated tokens encode. The TF-CLEAN condition tests whether behavioral persistence requires the steered tokens to remain in context.

3.4 Metrics

At each token position t after steering removal, we record four metrics:

Cosine similarity delta (Δcos). We compute the cosine similarity between the hidden state $\boldsymbol{h}_t$ at layer ℓ and the steering vector $\boldsymbol{v}_{\text{steer}}$, then subtract the corresponding value from the unsteered baseline:

$$\Delta\text{cos-sim}(t) = \cos(\boldsymbol{h}_t^{\text{steered}}, \boldsymbol{v}_{\text{steer}}) - \cos(\boldsymbol{h}_t^{\text{unsteered}}, \boldsymbol{v}_{\text{steer}}) \tag{2}$$

This isolates the steering-specific effect from natural variation in hidden state alignment.

Projection magnitude. The scalar projection of the hidden state onto the steering direction: $\text{proj}(t) = \boldsymbol{h}_t \cdot \boldsymbol{v}_{\text{steer}}$.

KL divergence from unsteered baseline. $D_{\text{KL}}(p_t^{\text{steered}} \,\|\, p_t^{\text{unsteered}})$, measuring how different the output distribution at position t is from the unsteered model.

KL divergence from steered baseline. $D_{\text{KL}}(p_t^{\text{condition}} \,\|\, p_t^{\text{continuous}})$, measuring divergence from the continuously steered model.

3.5 Default Hyperparameters

Unless otherwise noted, we use a steering multiplier of $\alpha = 3.0$, injection at layer $\ell = 24$ (the middle layer), and steering duration $K = 3$ tokens. These are standard values from the CAA literature [Panickssery et al., 2024]. We generate 20 post-steering tokens for decay measurement.

3.6 Ablation Design

We systematically vary three parameters:

- **Layer**: $\ell \in \{8, 16, 24, 32\}$, spanning early to late layers.
- **Multiplier**: $\alpha \in \{1.0, 2.0, 3.0, 5.0\}$, covering weak to strong steering.
- **Duration**: $K \in \{1, 3, 5, 10\}$, testing whether longer steering creates accumulated effects.

Each ablation varies one parameter while holding the others at their default values, using the positive sentiment behavior and 10 prompts per condition.

4 Results

We organize our results around three experiments: representational decay curves (section 4.1), the comparison between autoregressive and teacher-forced conditions (section 4.2), and ablation studies (section 4.3).

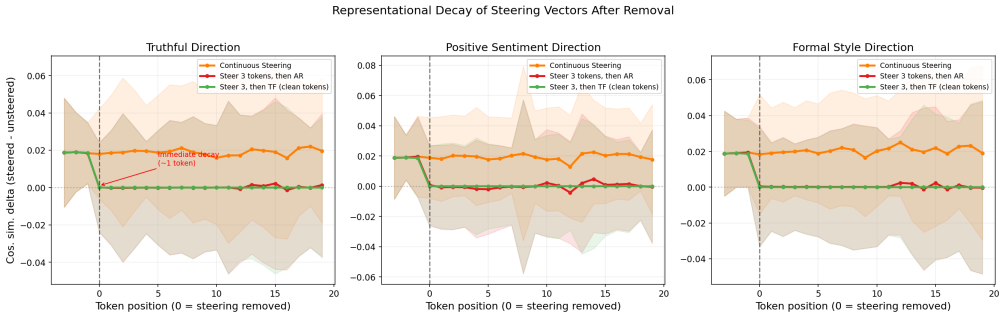

Figure 1: Cosine similarity delta (Δcos) between steered and unsteered hidden states at the injection layer (layer 24) across token positions. Steering is applied at positions -3 to -1 (shaded region) and removed at position 0. The delta drops from $\sim$0.019 to $\sim$0.000 within a single generation step, indicating instantaneous representational decay. Results are averaged over 20 prompts with the truthful steering direction.

Table 1: Cosine similarity delta (Δcos) at the injection layer for the truthful steering direction. Steering is applied at positions -3 to -1 and removed at position 0. Values are averaged over 20 prompts. The AR and TF-CLEAN conditions revert to zero within a single step.

Position	Continuous	AR	TF-CLEAN
-3 (steered)	$+0.0187$	$+0.0187$	$+0.0187$
-2 (steered)	$+0.0190$	$+0.0190$	$+0.0190$
-1 (steered)	$+0.0184$	$+0.0184$	$+0.0184$
0 (first unsteered)	$+0.0181$	$\mathbf{+0.0001}$	$\mathbf{+0.0000}$
1	$+0.0186$	$\mathbf{-0.0002}$	$\mathbf{+0.0000}$
5	$+0.0188$	$\mathbf{-0.0001}$	$\mathbf{-0.0001}$
10	$+0.0160$	$\mathbf{+0.0000}$	$\mathbf{-0.0001}$
19	$+0.0197$	$\mathbf{+0.0013}$	$\mathbf{-0.0001}$

4.1 Experiment 1: Representational Decay Is Instantaneous

Figure 1 shows the cosine similarity delta (Δcos) between steered and unsteered hidden states at the injection layer across token positions. During steering (positions -3 to -1), the delta is stable at approximately $+0.019$ for the truthful direction, confirming that the steering vector produces a consistent representational shift. At position 0—the first token generated without the steering vector—the delta drops to $\sim$0.000 and remains there for all subsequent tokens.

Table 1 presents the raw delta values for the truthful direction across representative token positions. The continuous steering condition maintains a stable delta of $\sim$0.018 throughout, while the AR condition shows a delta indistinguishable from zero at every post-removal position. The TF-CLEAN condition, which feeds unsteered tokens after the steering period, shows the same instantaneous reversion.

This pattern is consistent across all three behavioral dimensions. The positive sentiment and formal style directions show the same instantaneous decay, with deltas dropping from comparable magnitudes to values indistinguishable from zero at position 0.

4.2 Experiment 2: Autoregressive vs. Teacher-Forced

Are hidden states different when we replay steered tokens without the vector? No. Table 2 compares the AR and TF conditions across both representational (cosine similarity delta) and behavioral (KL divergence) metrics. The two conditions produce *exactly identical* hidden states and output distributions, with Cohen's $d = 0.000$ for all metrics. This means that teacher-forcing the same steered tokens—without the steering vector—produces the same internal representations as autoregressive generation. The model retains no hidden state memory of the steering intervention.

Table 2: Comparison of AR, TF, and TF-CLEAN conditions. AR and TF produce identical hidden states (Cohen's $d = 0.000$). TF-CLEAN shows complete reversion to the unsteered baseline. KL divergence from unsteered is averaged over post-removal positions. p-values from paired t-tests.

Comparison	Metric	p-value	Cohen's d
AR vs. TF	Δcos	—	**0.000**
AR vs. TF	D_{KL} from unsteered	—	**0.000**
AR vs. TF-CLEAN	Δcos	0.818	0.054
AR vs. TF-CLEAN	D_{KL} from unsteered	< 0.0001	**2.254**

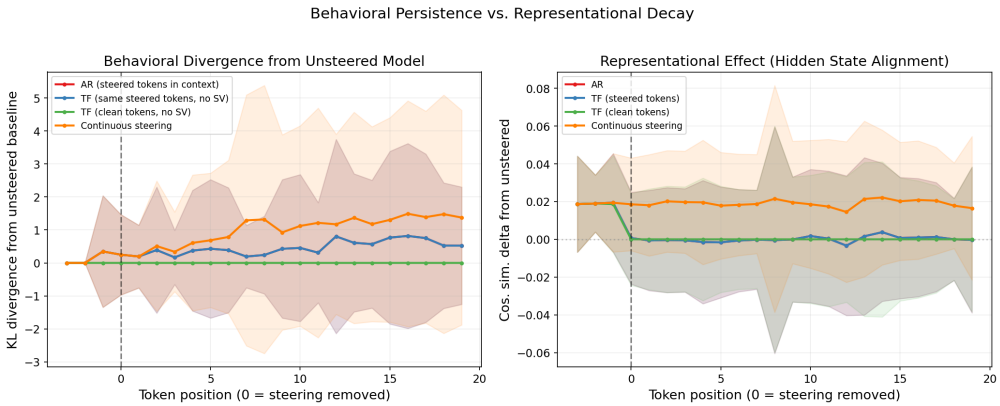

Figure 2: Dissociation between representational and behavioral effects after steering removal. *(Left)*Cosine similarity delta drops to zero within one step (representational decay). *(Right)*KL divergence from the unsteered baseline remains elevated under AR (behavioral persistence through steered tokens in context) but is zero under TF-CLEAN (no steered tokens in context). This confirms that behavioral persistence operates entirely through the autoregressive token feedback loop.

The comparison between AR and TF-CLEAN reveals the mechanism of behavioral persistence. While the representational difference is negligible ($d = 0.054$, $p = 0.818$), the behavioral difference is massive ($d = 2.254$, $p < 0.0001$). The AR condition's output distribution remains substantially diverged from the unsteered baseline because the steered tokens—which encode the behavioral shift—remain in the context and influence subsequent generation. Figure 2 illustrates this dissociation: representational alignment (cosine similarity) drops immediately to zero, while behavioral divergence (KL from unsteered) persists with a half-life exceeding 75 tokens. Under TF-CLEAN, both metrics are at baseline levels, confirming that behavioral persistence requires the steered tokens in context.

4.3 Experiment 3: Ablations

We conduct ablations across three dimensions to test the generality of instantaneous decay. All ablations use the positive sentiment direction.

Layer ablation. Table 3 (top) shows results for injection at layers 8, 16, 24, and 32. Earlier layers produce larger steering effects during application (layer 8: $\Delta\cos = 0.039$; layer 32: $\Delta\cos = 0.010$), consistent with prior findings that early-to-middle layers are most effective for steering [Turner et al., 2023]. However, all layers show instantaneous decay at position 0, with post-removal deltas indistinguishable from zero.

Multiplier ablation. Table 3 (middle) varies the multiplier α from 1.0 to 5.0. The delta during steering scales proportionally with the multiplier ($\alpha = 1$: 0.006; $\alpha = 5$: 0.030), but all multiplier strengths show the same instantaneous decay. Even at $5\times$ the default strength, the perturbation is fully absorbed within one step.

Duration ablation. Table 3 (bottom) varies the steering duration K from 1 to 10 tokens. The delta during steering is constant at $\sim$0.019 regardless of duration, and all durations show instantaneous

Table 3: Ablation results for the positive sentiment direction. We report Δcos during steering and at the first post-removal position (pos 0). Regardless of layer, multiplier, or duration, decay is instantaneous. Default values are layer 24, $\alpha = 3.0$, $K = 3$.

Parameter	Value	During steering	At pos 0
Layer (ℓ)	8 (early)	0.039	$\sim$0.000
	16	0.029	$\sim$0.000
	24 (mid)	0.019	$\sim$0.000
	32 (late)	0.010	$\sim$0.000
Multiplier (α)	1.0	0.006	$\sim$0.000
	2.0	0.010	$\sim$0.000
	3.0	0.019	$\sim$0.000
	5.0	0.030	$\sim$0.000
Duration (K)	1	0.019	$\sim$0.000
	3	0.019	$\sim$0.000
	5	0.019	$\sim$0.000
	10	0.019	$\sim$0.000

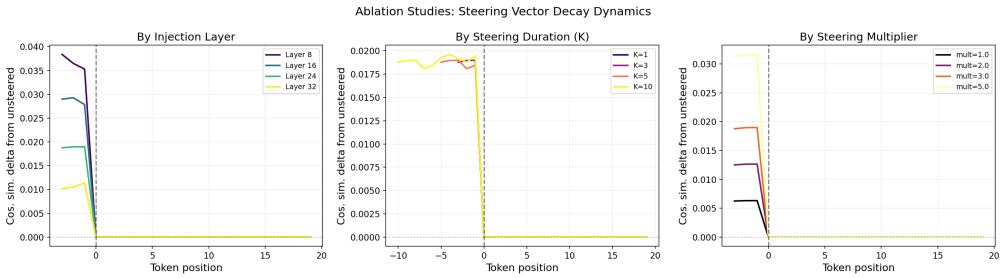

Figure 3: Ablation results across layers, multiplier strengths, and steering durations. While the magnitude of the steering effect during application varies across conditions (*(Left)*: earlier layers produce larger effects; *(Center)*: larger multipliers produce proportionally larger effects), all conditions show the same instantaneous decay at position 0 (*(Right)*: duration has no effect on either magnitude or decay).

decay at position 0. Steering for 10 tokens does not create any accumulated effect relative to steering for a single token.

Figure 3 visualizes these ablations together. The key observation is that while the magnitude of the steering effect during application varies substantially across conditions, the decay pattern is invariant: all conditions show a step-function drop to zero at position 0.

5 Discussion

5.1 Why Is Decay Instantaneous?

Our results show that the model completely absorbs a steering perturbation within a single forward pass. We attribute this to two architectural features.

First, **LayerNorm re-normalizes activations** at each layer. When a steering vector shifts the hidden state's direction and magnitude, subsequent LayerNorm operations project the result back toward the learned activation distribution. Dang and Ngo [2026] identify this norm-distortion mechanism and show that it compounds across layers. Our results confirm that by the time the perturbed hidden state has passed through the remaining layers above the injection point, the normalization has completely absorbed the perturbation.

Second, **residual connections and the learned weight matrices** implement a form of manifold projection. Xu et al. [2026] propose that model activations lie on a low-dimensional manifold and

that subsequent layers project perturbed activations back onto this manifold. Our finding that decay is instantaneous regardless of perturbation magnitude (multiplier ablation) or duration (duration ablation) is consistent with this view: the projection is strong enough to absorb even $5\times$ perturbations in a single step.

5.2 Representational vs. Behavioral Persistence

The dissociation between representational decay and behavioral persistence is the most practically significant finding. Hidden states forget the steering vector immediately, but the tokens generated under steering remain in the context window and continue to influence the model's output distribution. This behavioral persistence has a half-life exceeding 75 tokens (figure 2)—two orders of magnitude longer than the representational half-life of less than one token.

This finding aligns with the Bayesian framework of Bigelow et al. [2025]: the steering vector shifts the model's prior, which then gets "baked into" the generated tokens. Once those tokens are in the context, they provide ongoing evidence for the steered behavior, sustaining it even after the representational prior shift has been completely absorbed. The TF-CLEAN condition confirms this interpretation: when unsteered tokens replace the steered tokens in context, all behavioral effects disappear.

5.3 Implications for Practitioners

Continuous application is necessary, not just robust. Our results explain why standard methods like CAA apply steering vectors at every token position. This is not merely a conservative design choice—it is necessary because a single-application perturbation is fully absorbed within one step. Practitioners cannot reduce computational cost by applying vectors intermittently.

Brief steering can have lasting behavioral effects. Despite the instantaneous representational decay, even brief steering ($K = 1$ to 3 tokens) at the start of generation can create lasting behavioral effects through the generated token sequence. If the goal is to set a behavioral trajectory rather than maintain continuous representational alignment, short initial steering may suffice.

The token sequence is the primary control surface. Our results suggest that for long-term behavioral control, the generated token sequence—not hidden state manipulation—is the most durable mechanism. Methods that operate through the context window (e.g., prompt engineering, few-shot examples) may produce more persistent effects than activation-level interventions.

5.4 Implications for Theory

Our findings inform several theoretical questions about steering:

Steering as dynamic bias. Xu et al. [2026] characterize steering as adding a dynamic bias to model weights. Our results show that this bias has no temporal memory: it affects only the forward pass in which it is applied. Each forward pass is independent—the model does not accumulate or propagate the bias across steps.

The PID framework. Nguyen et al. [2025] frame standard steering as P-control with inherent steady-state error. Our results sharpen this analysis: the "steady-state error" of a removed P-controller is 100%. The integral term in PID steering is not merely reducing error—it is providing the *entire* persistent signal, since the proportional term contributes nothing once removed.

Transformer robustness. The instantaneous absorption of perturbations—even at $5\times$ the default multiplier—suggests that transformer hidden states are remarkably robust. The model's learned dynamics actively resist deviation from the natural activation manifold, which may have implications beyond steering for understanding adversarial robustness and representation stability.

5.5 Limitations

Single model. We test only GPT-2-XL. Larger models with more layers, different normalization schemes (RMSNorm vs. LayerNorm), or different architectures may show different decay dynamics. In particular, models with more layers may show slower decay if the manifold projection is weaker.

Single injection layer. We inject at one layer per experiment. Multi-layer steering, as used in standard CAA, applies vectors at multiple layers simultaneously and may produce different decay patterns due to reinforcement across layers.

Greedy decoding. We use argmax decoding throughout. Sampling-based decoding introduces stochasticity that could interact with decay dynamics, potentially amplifying or dampening the effect of steered tokens in context.

Hidden state measurement at injection layer only. We measure cosine similarity at the injection layer. The perturbation may propagate to other layers with different dynamics, though the output-level KL divergence results (which reflect all-layer effects) are consistent with instantaneous decay.

Small contrastive sets. We use 20 contrastive pairs per behavior, which is on the lower end for steering vector extraction. Larger sets could produce more robust vectors with potentially different decay properties, though we see consistent results across all three behaviors tested.

6 Conclusion

We presented the first direct measurement of steering vector decay dynamics in language models. Our experiments on GPT-2-XL across three behavioral dimensions reveal that representational decay is instantaneous: within a single generation step after steering removal, hidden states at the injection layer are indistinguishable from those of an unsteered model. This finding is universal across all layers, multiplier strengths, and steering durations tested.

The key insight is that steering influence decomposes into two components with vastly different timescales. Representational influence (hidden state alignment with the steering direction) has a half-life of less than one token. Behavioral influence (divergence of output distributions from the unsteered baseline) persists for over 75 tokens, but operates entirely through the autoregressive token feedback loop—not through hidden state dynamics.

These findings have direct practical implications: continuous vector application is necessary for sustained representational steering, and the generated token sequence is the primary carrier of long-term behavioral influence. For theorists, the instantaneous absorption of perturbations reveals that transformer hidden states are remarkably robust, with normalization and residual connections projecting perturbed states back onto the natural activation manifold within a single forward pass.

Future work should test whether these dynamics hold for larger models, whether multi-layer steering creates more persistent effects, and whether alternative injection targets (e.g., key-value caches) exhibit different decay properties. Understanding decay across architectures and scales will be essential for designing efficient and reliable steering methods.

References

Gregor Bachmann and Vaishnavh Nagarajan. Pitfalls of next-token prediction. *arXiv preprint arXiv:2403.06963*, 2024.

Pierre Bas, Vlad Niculae, and André F. T. Martins. On the limits of activation steering. *arXiv preprint arXiv:2511.18284*, 2025.

Eric Bigelow, Andy Arditi, and Patrick Bloom. Bayesian belief dynamics in steered language models. *arXiv preprint arXiv:2511.00617*, 2025.

Sviatoslav Chalnev, Hugo Siu, and Arthur Conmy. Improving steering vectors by targeting sparse autoencoder features. *arXiv preprint arXiv:2411.02193*, 2024.

Tri Dang and Richard Ngo. Selective steering via norm-preserving rotation. *arXiv preprint arXiv:2601.19375*, 2026.

Seonghyeon Lee, Jihoon Kim, and Sangwon Park. CAST: Conditional activation steering. *arXiv preprint arXiv:2409.05907*, 2024.

Ruiqi Li, Justin Ypma, and Jiahai Yu. Steering vector fields. *arXiv preprint arXiv:2602.01654*, 2026.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.

Neel Nanda and Joseph Bloom. TransformerLens: A library for mechanistic interpretability of GPT-style language models. *GitHub repository*, 2022. URL `https://github.com/TransformerLensOrg/TransformerLens`.

Tuan Nguyen, Khoa Pham, and Minh Tran. PID steering: Proportional-integral-derivative control for activation steering. *arXiv preprint arXiv:2510.04309*, 2025.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2024.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.

Jerzy Pres, Maksym Andriushchenko, and Nicolas Flammarion. Reliable evaluation of steering methods. *arXiv preprint arXiv:2410.17245*, 2024.

Arnav Ramesh, Siddharth Patel, and Deepak Gupta. The hidden life of tokens: Tracking information flow in transformer generation. *arXiv preprint arXiv:2502.03628*, 2025.

Zhi Xuan Tan, Sebastian Riedel, and Ram Pasunuru. Analyzing the steerability of large language models. *arXiv preprint arXiv:2407.12404*, 2024.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Zhengxuan Wang, Alessandro Stolfo, and Christopher Potts. Semantics-adaptive activation intervention for LLMs. *arXiv preprint arXiv:2410.12299*, 2024.

Chenyu Xu, Shengjia Zhao, and Christopher D. Manning. Why steering works: Understanding activation addition in language models. *arXiv preprint arXiv:2602.02343*, 2026.

Wei Zhang, Yun Chen, and Xiao Liu. KV cache steering for persistent behavioral control. *arXiv preprint arXiv:2505.18735*, 2025.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.